

SCX 5

O(d) Cartan vs.

Viewpoint 5 Correction: Cartan Linearization vs. Full Non-Abelian Gauge-Fixing

SCX

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1 () Original Claim (Under Review)

5 ():

ACE O(3) O(d) O(d) (non-abelian gauge fixing) Riemannian Gauss-Newton

Viewpoint 5 (Original):

ACE potentials possess O(3) rotational symmetry, making the gauge group O(d) rather than the translation group. Because O(d) is a non-abelian Lie group, gauge fixing requires **non-abelian gauge fixing** techniques, implemented as Riemannian Gauss-Newton optimization on the manifold. This procedure has quadratic convergence.

2 Review Verdict

: **(OVERSTATED)**

- (1) O(d)
- (2) Cartan $\mathfrak{so}(d)$
- (3) “” Yang-Mills
Gribov —
- (4) Gauss-Newton

Verdict: OVERSTATED

The original claim is partially correct in its technical content, but its formulation is significantly overstated:

- (1) It conflates the group-theoretic non-abelian property of O(d) with the need for full non-abelian gauge-fixing machinery;
- (2) It overlooks the critical fact that Cartan decomposition linearizes the problem to the Lie algebra $\mathfrak{so}(d)$;
- (3) “Non-abelian gauge fixing” in physics specifically denotes Yang-Mills techniques — ghost fields, Gribov copies, etc. — **none of which apply**;
- (4) Gauss-Newton has only **linear convergence** (not quadratic) when residuals are non-zero.

3

Why the Original Claim Is Overstated

3.1 $O(d)$

$O(d)$ Being Non-Abelian $\neq$ Requiring Non-Abelian Gauge-Fixing

. $O(d)$ $R_1, R_2 \in O(d)$ $R_1 R_2 \neq R_2 R_1$ $O(d)$ $\mathfrak{so}(d)$ $\frac{d(d-1)}{2}$ Cartan

English. $O(d)$ is indeed non-abelian as a group: for $R_1, R_2 \in O(d)$, generally $R_1 R_2 \neq R_2 R_1$. However, **the algebraic property of the group and the algorithmic complexity of solving the gauge-fixing problem are distinct concerns.** The crucial fact is: $O(d)$ as a Lie group has Lie algebra $\mathfrak{so}(d)$, which is a **vector space** (of dimension $\frac{d(d-1)}{2}$), and Cartan decomposition — the inverse of the exponential map, i.e., the logarithm — provides a local diffeomorphism from the group manifold to this vector space.

3.2 “”

What “Non-Abelian Gauge Fixing” Means in Physics

. “” Yang-Mills Faddeev-Popov

- (1) **(Ghost Fields):**
- (2) **Gribov (Gribov Copies):** —
- (3) **BRST :**
- (4) **: Jacobian Faddeev-Popov**

ACE $O(d)$ ACE

English. In quantum field theory, “non-abelian gauge fixing” (e.g., the Faddeev-Popov procedure in Yang-Mills theory) involves:

- (1) **Ghost Fields:** Anti-commuting scalar fields introduced to cancel unphysical gauge degrees of freedom;
- (2) **Gribov Copies:** Non-uniqueness of gauge-fixing conditions — the same physical configuration may correspond to multiple gauge-fixed solutions;
- (3) **BRST Symmetry:** A super-symmetry structure relating ghosts and gauge fields;
- (4) **Path-Integral Measure:** A Jacobian determinant (Faddeev-Popov determinant) is required to correct the integration measure.

None of these issues apply to the $O(d)$ gauge-fixing problem for ACE potentials. The ACE gauge-fixing is a finite-dimensional parameter estimation problem, not an infinite-dimensional quantum field theory path-integral problem.

3.3 Gauss-Newton

Convergence Order of Gauss-Newton

. Gauss-Newton

- Gauss-Newton

- ACE — Gauss-Newton
- $\|x_{k+1} - x^*\| \leq \rho \|x_k - x^*\|$ $\rho < 1$

English. The original claim asserts quadratic convergence for Gauss-Newton. This misreads standard numerical optimization theory:

- Gauss-Newton exhibits **local quadratic convergence** only when **residuals are zero** (zero-residual problem, where the problem is exactly satisfied at the solution);
- When residuals are non-zero (the typical case for ACE gauge-fixing — there is no exact solution that perfectly eliminates all gauge redundancy), Gauss-Newton has only **linear convergence**;
- The linear convergence rate is governed by the problem's condition number: $\|x_{k+1} - x^*\| \leq \rho \|x_k - x^*\|$, where $\rho < 1$ but ρ may be close to 1.

4

Corrected Claim

1 (5). ACE $O(3)$ $O(d)$ $O(d)$ **Cartan** $g_m = g_0 \cdot \exp(\xi_m)$ $\xi_m \in \mathfrak{so}(d)$ $\mathfrak{so}(d)$ $\frac{d(d-1)}{2}$
 $\mathcal{O}(d^2)$ Riemannian Gauss-Newton

For applications where the log map is valid (near the identity), this linearization adds only $\mathcal{O}(d^2)$ overhead. The full Riemannian Gauss-Newton is needed only when far from the solution, and converges **linearly** (not quadratically) when residuals are non-zero. *ACE potentials possess $O(3)$ rotational symmetry, making the gauge group $O(d)$ rather than the translation group. While $O(d)$ is non-abelian as a group, the gauge-fixing problem can be linearized via **Cartan decomposition**: writing $g_m = g_0 \cdot \exp(\xi_m)$ with $\xi_m \in \mathfrak{so}(d)$ in the Lie algebra reduces it to a **standard linear least-squares problem** on a vector space of dimension $\frac{d(d-1)}{2}$.*

*For applications where the log map is valid (near the identity), this linearization adds only $\mathcal{O}(d^2)$ overhead. The full Riemannian Gauss-Newton is needed only when far from the solution, and converges **linearly** (not quadratically) when residuals are non-zero.*

5

Mathematical Justification

5.1 Cartan

Cartan Decomposition and the Logarithm Map

Definition 2 (Cartan). $G = O(d)$ d

$$\mathfrak{so}(d) = \{\Omega \in \mathbb{R}^{d \times d} : \Omega^\top = -\Omega\}, \quad (1)$$

$$d \times d \mathfrak{so}(d) \frac{d(d-1)}{2}$$

Cartan $g \in O(d)$

$$g = \exp(\xi), \quad \xi \in \mathfrak{so}(d), \quad (2)$$

$$\exp : \mathfrak{so}(d) \rightarrow O(d) \quad \log : O(d) \rightarrow \mathfrak{so}(d) \quad O(d) \quad \det(g) = -1 \quad g \rightarrow -1$$

Definition (Cartan Decomposition). Let $G = O(d)$ be the d -dimensional orthogonal group with Lie algebra:

$$\mathfrak{so}(d) = \{\Omega \in \mathbb{R}^{d \times d} : \Omega^\top = -\Omega\}, \quad (3)$$

the set of all $d \times d$ skew-symmetric matrices. $\mathfrak{so}(d)$ is a **vector space** of dimension $\frac{d(d-1)}{2}$.

Cartan decomposition states: every $g \in O(d)$ near the identity can be uniquely written as:

$$g = \exp(\xi), \quad \xi \in \mathfrak{so}(d), \quad (4)$$

where $\exp : \mathfrak{so}(d) \rightarrow O(d)$ is the matrix exponential. Its inverse $\log : O(d) \rightarrow \mathfrak{so}(d)$ (the matrix logarithm) is defined everywhere on $O(d)$ except a measure-zero subset where $\det(g) = -1$ and g has eigenvalue -1 .

5.2

Linearization of the Gauge-Fixing Problem

• ACE $\{g_m\}_{m=1}^M \subset O(d)$ M $\mathcal{L}(\{g_m\})$

1: $g_0 \in O(d)$

2: g_m

$$g_m = g_0 \cdot \exp(\xi_m), \quad \xi_m \in \mathfrak{so}(d). \quad (5)$$

3: $\xi_m \in \mathfrak{so}(d) \cong \mathbb{R}^{d(d-1)/2}$ $\{E_{ij} = e_i e_j^\top - e_j e_i^\top\}_{1 \leq i < j \leq d}$

$$\xi_m = \sum_{1 \leq i < j \leq d} \omega_m^{ij} \cdot E_{ij}, \quad \omega_m^{ij} \in \mathbb{R}. \quad (6)$$

4: $\mathcal{L} \{\omega_m^{ij}\}$

$$\mathcal{L}(\{\omega_m^{ij}\}) \approx \mathcal{L}_0 + \mathbf{J} \cdot \boldsymbol{\omega} + \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{H} \boldsymbol{\omega}, \quad (7)$$

$\boldsymbol{\omega} \in \mathbb{R}^{M \cdot d(d-1)/2}$ $\mathbf{J}$ Jacobian $\mathbf{H}$ Hessian

5: $\mathbf{H} \cdot \boldsymbol{\omega}^* = -\mathbf{J}^\top$ Newton exp

2-5 $\mathbb{R}^{M \cdot d(d-1)/2}$ $O(d)^M$ Cartan ———

English. Consider the gauge-fixing problem in ACE potentials. Let $\{g_m\}_{m=1}^M \subset O(d)$ be the gauge transformations (rotation matrices) of M experts. The gauge-fixing objective is to minimize a loss function $\mathcal{L}(\{g_m\})$ measuring inconsistency among rotated potentials.

Linearization procedure:

Step 1: Choose a reference rotation $g_0 \in O(d)$ (e.g., the current estimate).

Step 2: Parameterize each g_m as:

$$g_m = g_0 \cdot \exp(\xi_m), \quad \xi_m \in \mathfrak{so}(d). \quad (8)$$

Step 3: Vectorize the skew-symmetric matrix ξ_m . $\mathfrak{so}(d)$ is isomorphic to $\mathbb{R}^{d(d-1)/2}$ with basis $\{E_{ij} = e_i e_j^\top - e_j e_i^\top\}_{1 \leq i < j \leq d}$:

$$\xi_m = \sum_{1 \leq i < j \leq d} \omega_m^{ij} \cdot E_{ij}, \quad \omega_m^{ij} \in \mathbb{R}. \quad (9)$$

Step 4: Expand the loss $\mathcal{L}$ to first order in $\{\omega_m^{ij}\}$:

$$\mathcal{L}(\{\omega_m^{ij}\}) \approx \mathcal{L}_0 + \mathbf{J} \cdot \boldsymbol{\omega} + \frac{1}{2} \boldsymbol{\omega}^\top \mathbf{H} \boldsymbol{\omega}, \quad (10)$$

where $\boldsymbol{\omega} \in \mathbb{R}^{M \cdot d(d-1)/2}$ concatenates all parameters, $\mathbf{J}$ is the gradient (Jacobian), and $\mathbf{H}$ is the approximate Hessian.

Step 5: Solve the linear system $\mathbf{H} \cdot \boldsymbol{\omega}^* = -\mathbf{J}^\top$ for the Newton step, then map back to the group manifold via $\exp$.

Key insight: All computations in Steps 2–5 take place in the **vector space** $\mathbb{R}^{M \cdot d(d-1)/2}$, not on the curved group manifold $O(d)^M$. This is the power of Cartan decomposition — transforming an optimization problem on a non-abelian group into a standard linear least-squares problem on a vector space.

5.3

Complexity Analysis

Proposition 3. $M \ d$

- $M \cdot \frac{d(d-1)}{2}$
 - $\exp(\xi_m) \ \mathcal{O}(d^3) \text{ Rodrigues } \mathcal{O}(d^2) \ d = 3 \ SO(3)$
 - *Jacobian* $\mathcal{O}(M \cdot d^3)$
 - $\mathcal{O}((Md^2)^3) \ \mathcal{O}(Md^2 \cdot \kappa)$
- $(d = 3, M \lesssim 100) \ \mathcal{O}(M \cdot 27) \text{ ——— Riemannian}$

Proposition (Computational Overhead of Linearization). For M experts and d -dimensional rotation group:

- Total parameters: $M \cdot \frac{d(d-1)}{2}$.
- Matrix exponential $\exp(\xi_m)$ per iteration: $\mathcal{O}(d^3)$ (reducible to $\mathcal{O}(d^2)$ via Rodrigues formula, especially for $d = 3$ i.e., $SO(3)$).
- Jacobian computation: $\mathcal{O}(M \cdot d^3)$ via automatic differentiation or analytic gradients.
- Linear system solve: $\mathcal{O}((Md^2)^3)$ for dense direct solver, or $\mathcal{O}(Md^2 \cdot \kappa)$ for conjugate gradient.

For practical settings ($d = 3, M \lesssim 100$), total overhead is $\mathcal{O}(M \cdot 27)$ floating-point operations — entirely feasible and far cheaper than full Riemannian optimization.

6 Riemannian Gauss-Newton

When Riemannian Gauss-Newton IS Needed

. Cartan $\log : O(d) \rightarrow \mathfrak{so}(d)$

$$(1) : g_m > \pi/2 \log(g_m)$$

$$(2) : O(d) \det = -1$$

(3) : $O(d)$ Riemannian Riemannian trust-region

Riemannian Gauss-Newton ———
Riemannian

English. Cartan linearization is not a panacea. The logarithm map $\log : O(d) \rightarrow \mathfrak{so}(d)$ is valid only under:

- (1) **Proximity to identity:** When g_m is far from the identity (e.g., rotation angle $> \pi/2$), $\log(g_m)$ may not exist or may incur large linearization error;
- (2) **Reflection components:** $O(d)$ includes the $\det = -1$ reflection branch, where the logarithm has singularities;
- (3) **Large-scale global optimization:** When searching for a global optimum over all of $O(d)$, Riemannian optimization (e.g., Riemannian trust-region methods) is more robust.

In these cases, **Riemannian Gauss-Newton is the appropriate tool** — but its convergence remains linear (when residuals are non-zero), and its per-iteration cost far exceeds that of the linearized method (requiring computation of Riemannian connections, parallel transport, and geodesics).

7 Before-and-After Comparison

Table 1: 5Comparison Before and After Correction

Aspect	Original	Corrected
Gauge-fixing method	“” “Non-abelian gauge fixing”	Cartan $\rightarrow$ Cartan decomp. $\rightarrow$ vector-space linear LS
Mathematical mechanism	Yang-Mills Gribov Yang-Mills style (ghosts, Gribov)	$\log : O(d) \rightarrow \mathfrak{so}(d)$ Logarithm map $\log : O(d) \rightarrow \mathfrak{so}(d)$
Computational complexity	Unspecified, implied high	$\mathcal{O}(d^2)$ $\mathcal{O}(d^2)$ overhead, near-zero near identity
Gauss-Newton Gauss-Newton convergence	Claimed quadratic	Linear convergence (non-zero residuals)
Scope of applicability	Unqualified	Riemannian Linearized near identity; Riemannian when far

8 Summary

:
5——ACE $O(3)$ $O(d)$ —— “”

(1) **Cartan** : $\log : O(d) \rightarrow \mathfrak{so}(d)$ $\mathfrak{so}(d) \cong \mathbb{R}^{d(d-1)/2}$ —Gribov BRST

(2) : “” Faddeev-Popov ACE

(3) : Gauss-Newton ACE

Cartan $\mathcal{O}(d^2)$ Riemannian

English Summary:

The core physical intuition of Viewpoint 5 — that ACE potentials have $O(3)$ rotational symmetry, making the gauge group $O(d)$ — is **correct**. However, concluding that this requires “non-abelian gauge fixing” is **overstated** for three reasons:

- (1) **Existence of Cartan decomposition:** The logarithm map $\log : O(d) \rightarrow \mathfrak{so}(d)$ linearizes the problem on the group manifold to the vector space $\mathfrak{so}(d) \cong \mathbb{R}^{d(d-1)/2}$, reducing it to standard linear least squares — no ghost fields, Gribov copies, or BRST symmetry required.
- (2) **Misuse of physics terminology:** “Non-abelian gauge fixing” has a precise technical meaning in quantum field theory (Faddeev-Popov procedure), and none of that machinery applies to the finite-dimensional parameter estimation problem of ACE potentials.
- (3) **Incorrect convergence claim:** Gauss-Newton has quadratic convergence only in the zero-residual case. The ACE gauge-fixing problem has non-zero residuals, so convergence is only linear.

The corrected formulation accurately reflects the true mathematical structure: a non-abelian symmetry that can be linearized via Cartan decomposition, with $\mathcal{O}(d^2)$ solution complexity, requiring full Riemannian optimization only when far from the solution.

Acknowledgments

$O(d)$ Gauss-Newton

We thank the reviewer for the penetrating insight into the relationship between the $O(d)$ gauge group and its Lie algebra, and for the accurate reminder about Gauss-Newton convergence theory. This correction is based on standard results in numerical optimization, Lie group theory, and gauge fixing in quantum field theory.